

MECHANICAL DESIGN OF THE IN-VACUUM TAPERED UNDULATOR AT TAIWAN PHOTON SOURCE

W. Y. Lai, K. H. Hsu, C. K. Kuan

National Synchrotron Radiation Research Center, Hsinchu, Taiwan

Abstract

Synchrotron light sources commonly provide users with two types of insertion devices for experiments in biology, medicine, and other fields: in-vacuum undulators (IUs), which have short period lengths and are used for medium-energy photon generation, and cryogenic permanent magnet undulators (CPMUs), which are designed for higher-energy photons. The strong magnetic fields generated by these devices exert significant forces on the magnets, resulting in structural deformation that can degrade the magnetic field quality. This paper presents the design and measurement methods of an in-vacuum tapered undulator, analyzes both simulation and experimental results of its structural deformation, and demonstrates how a flexible structure can be employed to achieve nonlinear magnetic force compensation, thereby improving overall system performance.

INTRODUCTION

The in-vacuum tapered undulator (IUT) is a critical device in third-generation synchrotron light sources for generating hard X-rays[1,2]. To achieve hard X-ray production, a smaller undulator gap is required, which significantly increases the magnetic force. This force acts on the undulator's frame and crossbeams, leading to considerable structural deformation. Even after precise calibration, such deformations can cause subtle positional shifts in the magnets, resulting in variations in the magnetic field as the gap size changes. To address this, it is essential to design a rigid frame and crossbeam structure capable of limiting deformation to less than 10 μm under load. In addition to ensuring structural rigidity, this paper also presents the mechanical design of the in-vacuum tapered undulator and introduces a newly implemented flexure mechanism for compensating magnetic forces.

Requirements

The IUT mechanical system must meet six essential functional requirements:

- **Travel Range:**
The opening/closing motion must exceed 35 mm to accommodate energy tuning and magnets installation, with a usable magnet gap range of 5 mm to over 40 mm.
- **Tilt Range:**
In addition to vertical movement, the IUT requires tilt adjustment functionality. The required tilt angle adjustment range must exceed ± 0.5 mrad.
- **Resolution:**

The magnet gap must be adjustable with a resolution finer than 1 μm to meet the needs of magnetic field measurements and future beamline energy tuning.

- **Deformation Tolerance:**
The crossbeam must withstand a uniform load of 50 kN, with deformation under 5 μm . Additionally, the main frame of the undulator should ensure that crossbeam rotation under a 50 kN load is less than 0.1 mrad.
- **Weight Limitation:**
The total weight of the undulator system must be less than 13 metric tons, accounting for a 15-ton crane capacity minus 2 tons for the magnet and vacuum chamber.
- **Space Constraints:**
Each crossbeam should be 2 meters long to accommodate the magnet array. The overall system height must be under 2.5 meters, and the distance from the electron beam center to the aisle-side structure must be less than 1 meter.

Based on the above requirements, the IUT system must include two magnet-supporting crossbeams capable of both angular adjustment and opening/closing operations. Each 2-meter-long crossbeam must withstand a 50 kN load with deformation less than 10 μm . The design adopts a dual rectangular shaped crossbeam configuration to support the magnetic force. Each crossbeam is equipped with a rotational axis to enable tilting. Thus, at least two independent motion systems are required: one for opening/closing the beam and another for adjusting the beam angle, as illustrated in Fig. 1.

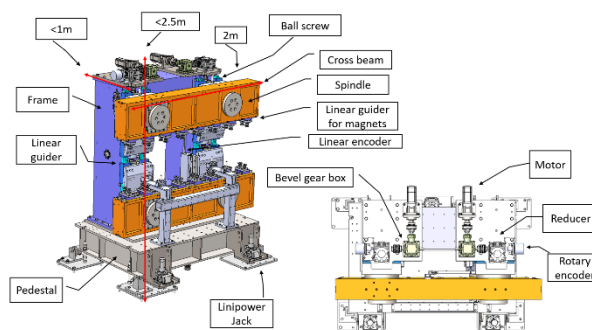


Figure 1: IUT Structure diagram.

Mechanism Design

The two primary deformation regions of the insertion device (ID) magnet are the cross beams that suspend the magnets and the main supporting frame that controls the opening and closing of these beams. The magnetic forces act on the magnet array beams, causing deformation that affects

the resulting magnetic field. The upper and lower beams are designed with mechanisms for angular adjustment and opening/closing, and they are fixed to two rotational shafts. These shafts are guided by linear rails to ensure straight-line motion. However, the magnetic force is transmitted through the linear rails to the main supporting frame, leading to deformation in the frame. This deformation subsequently alters the positions of the rotational shafts and cross beams.

Using simulation software (SolidWorks), the required structural strength can be determined. In the beam design, the optimal distance between the two rotational shafts was first established. Simulation analysis revealed that a spacing of 1100 mm between the shafts yields optimal performance (as shown in Fig. 2). Therefore, the shaft-to-shaft distance of the IUT (Inclined Undulator Type) insertion device magnet was set to 1100 mm.

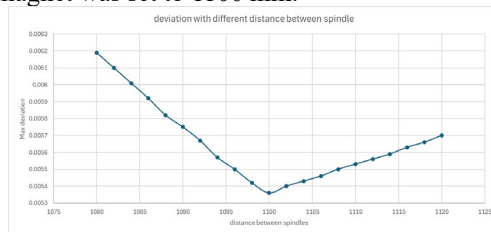


Figure 2: Optimal deformation curve.

To minimize deformation under load, increasing the moment of inertia is essential. Among the factors influencing the moment of inertia, beam height plays a critical role. Considering spatial constraints in the design, the beam height for the IUT was set at 420 mm. The rib thickness of the beam is 23 mm, and the material used is SS400 steel. In the simulation, the beam was subjected to a uniformly distributed load of 50 kN, with the shaft holes set as fixed boundary conditions (as shown in Fig. 3). The results showed that the maximum overall deformation was approximately 5 μm , and the relative height difference on the bottom surface was about 1.5 μm .

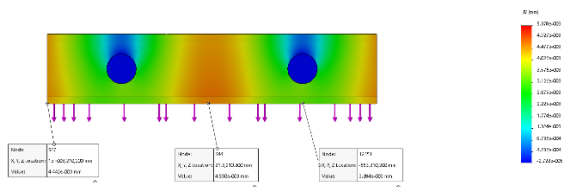


Figure 3: Cross beam deformation simulation results.

In addition to the primary deformation of the cross beams, the deformation of the main frame is another critical factor that cannot be ignored. If the frame lacks sufficient strength, the resulting deformation under load can lead to excessive Rz angular displacement, as shown in the figure below. Simulation results indicate that the relative angular change between the upper and lower cross beams is 0.061 mrad, which meets the required specification of 0.1 mrad.

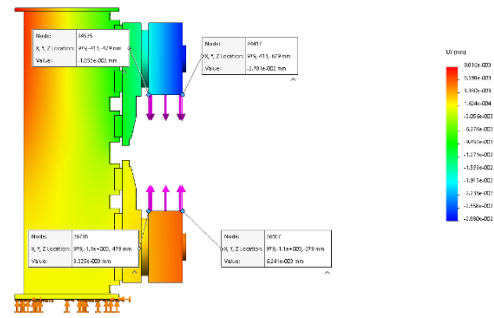


Figure 4: Frame deformation simulation results.

Compensating Mechanism

After the magnet is installed, when the gap is adjusted from 40 mm to 5 mm without feedback control, measurement results show that the movement difference between the two spindles reaches 35 μm . This clearly indicates that the two ball screws deform differently under load as shown in Fig. 5. The difference is glow up seriously after the gap smaller than 7.5 mm.

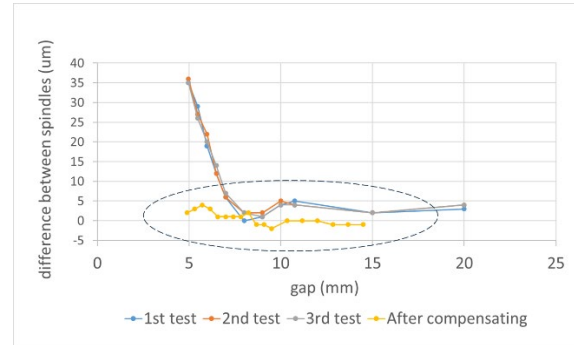


Figure 5: Variation with changing magnet gap.

From the theoretical magnetic field values, we know that when the magnet gap is 7.5 mm, approximately 400 kg of force is applied to the screw. Therefore, we designed a deformable, nonlinear force-compensation mechanism to counteract this magnetic force, reducing the force to below 200 kg after compensation.

The structure of the sheet module consists of four flexible plates, manufactured as a single integrated piece. The design principle is based on the concept that stiffness decreases when springs are connected in series. (as shown in Fig. 6)

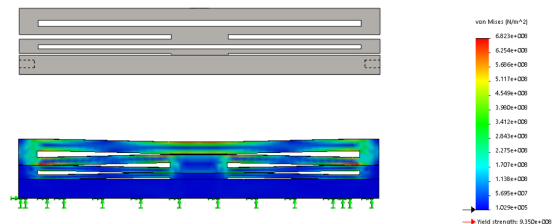


Figure 6: Diagram of the plate structure.

When no external force is applied, the structure behaves as four plates in series, exhibiting low stiffness. As the external force increases, the plate with the lowest stiffness will first come into contact with the base, effectively becoming part of the base structure and ceasing to deform. The continuing external force is then applied to the remaining three plates, thus increasing the overall stiffness and generating nonlinear resistance. Ultimately, only the top thick plate remains active, and the module provides its maximum compensating stiffness, which is limited by the strength of the topmost plate. The gap between each plate determines the displacement range at which each stiffness stage engages.

From the theoretical magnetic field data, we can see that the magnetic force increases exponentially as the gap decreases. Therefore, in the design, the structural strength must be progressively increased stage by stage in order to generate a compensating force with a similar exponential profile. Table 1 is used to design the required compensating force.

- **Column A** It is the required structural stiffness:
- **Column B** Specifies the allowable deformation gap between plates
- **Column F** The force required when each plate reaches its maximum deformation is calculated as column A $\times$ column B.
- **Column D** The stiffness for each stage is calculated using the formula for springs in series.

$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3} + \frac{1}{K_4} + \dots,$$
- **Column C** The sum stroke at each stage is obtained by Δ column F/ column C.
- **Column E** Set the starting point of the magnet gap, and subtract column C from this starting point.

Table 1: Reaction Load Design Table

	A	B	C	D	E	F
	[kgw/mm]	[mm]	[mm]	[kgw/mm]	[mm]	[kgw]
	stiffness	gap(ME)	stroke	K	gap(MG)	Max force
0th stage	0	0	0	0	14	0
1st stage	50	2.0	3.4	29.8	10.6	100
2nd stage	120	2.4	5.9	74.0	8.1	288
3rd stage	300	2.5	8.3	192.9	5.7	750
4th stage	540	3.1	10.0	540.0	4.0	1674

By adjusting the stiffness (Column A) and structural gap (Column B) in this table, we can obtain the designed force (Column F) and gap (Column E) curves. These designed curves are then compared with the theoretical magnetic field curve to ensure that the difference between the two is within the required limit of 200 kgf.

For the structural stiffness design, we used SolidWorks simulations to determine the required plate thicknesses. To verify the accuracy of the design, after manufacturing the force-compensation structure, we used a load cell and a

length gauge (as shown in Fig. 7) to measure and confirm whether the displacement–force curve matches the simulation results.

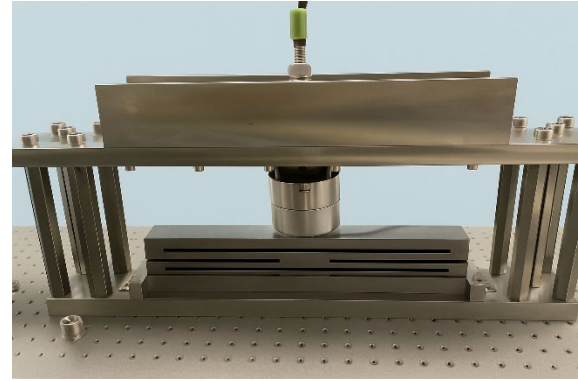


Figure 7: Photo of the experimental setup.

We plotted the theoretical magnetic force curve, the designed force curve, the simulated curve of the mechanism, and the experimentally validated curve, as shown in Fig. 8. The results indicate that the maximum deviation is approximately 100 kgf, meeting the required specifications. Finally, the movement difference between the two ball screws during gap adjustment from 40 mm to 5 mm—under open-loop operation—is significantly reduced to 4 μ m.

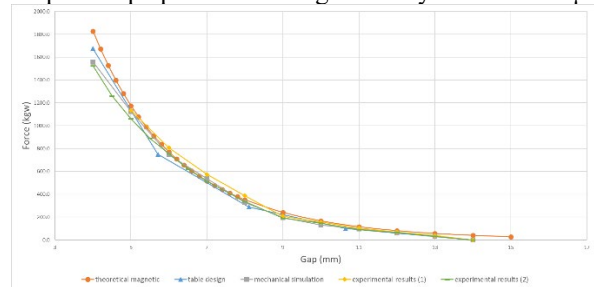


Figure 8: Results comparison graph.

CONCLUSION

This study presents the complete mechanical design, simulation analysis, and experimental validation of an invacuum tapered undulator. The undulator must operate under strong magnetic forces while keeping structural deformation within acceptable limits to ensure stable and high-quality magnetic fields. To address the nonlinear increase in magnetic force as the magnet gap changes, a novel flexure-based compensation mechanism was introduced. This mechanism provides nonlinear stiffness that matches the exponential growth of the magnetic force, effectively reducing mechanical imbalance.

REFERENCES

- [1] J. C. Huang *et al.*, “A mechanical undulator frame to minimize intrinsic phase errors”, in *Proc. MEDSI'18*, Paris, France, Jun. 2018, pp. 69-71.
doi:10.18429/JACoW-MEDSI2018-TUPH19
- [2] K. H. Hsu *et al.*, “Assembly process and inspection results for W100”, in *Proc. IPAC'24*, Nashville, TN, USA, May 2024, pp. 3788-3790.
doi:10.18429/JACoW-IPAC2024-THPS24